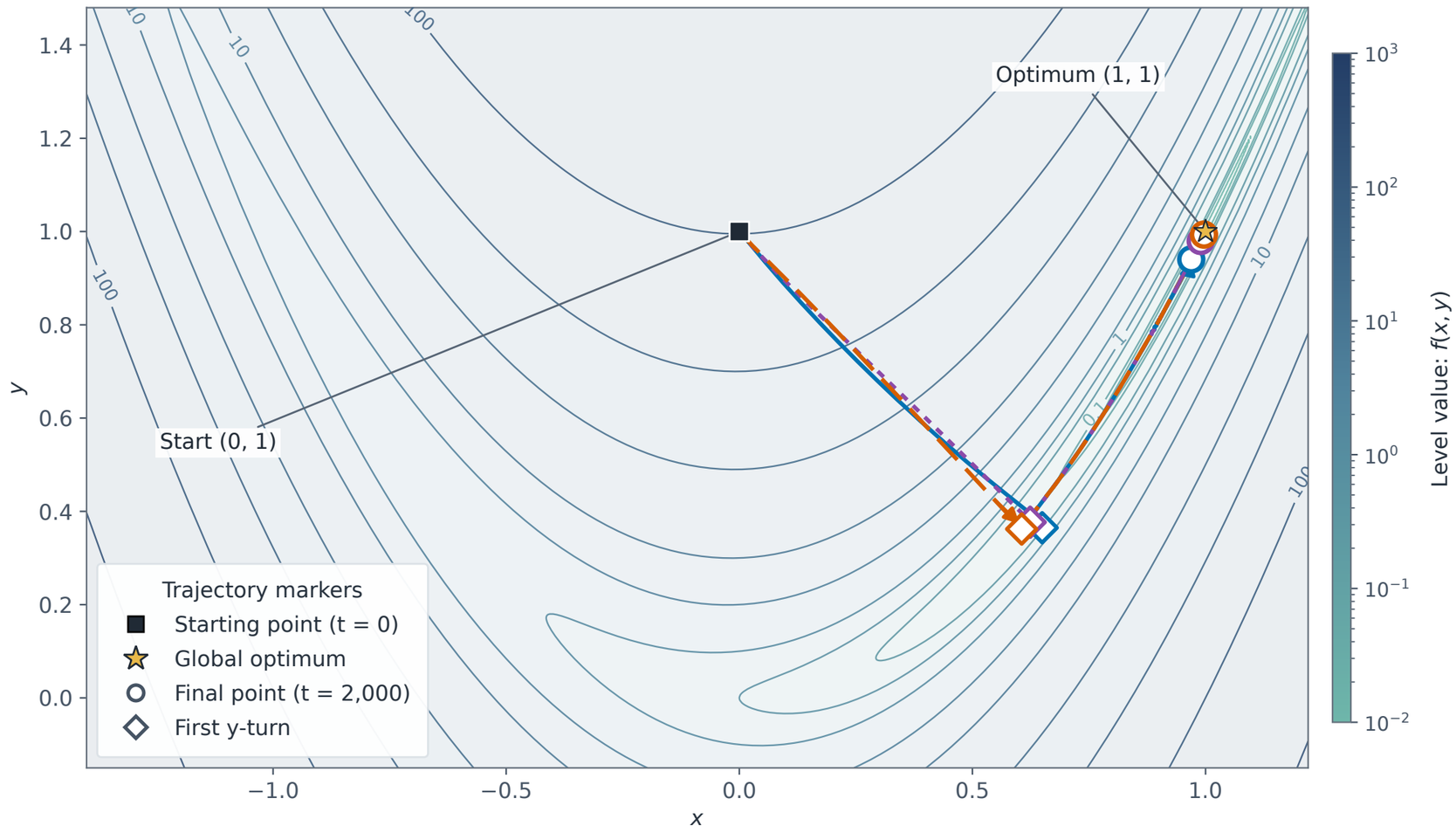
Rosenbrock · AdamW with momentum, weight decay 0.1

2,000 updates per learning rate · $\beta_1 = 0.9$, $\beta_2 = 0.95$ · CPU float64

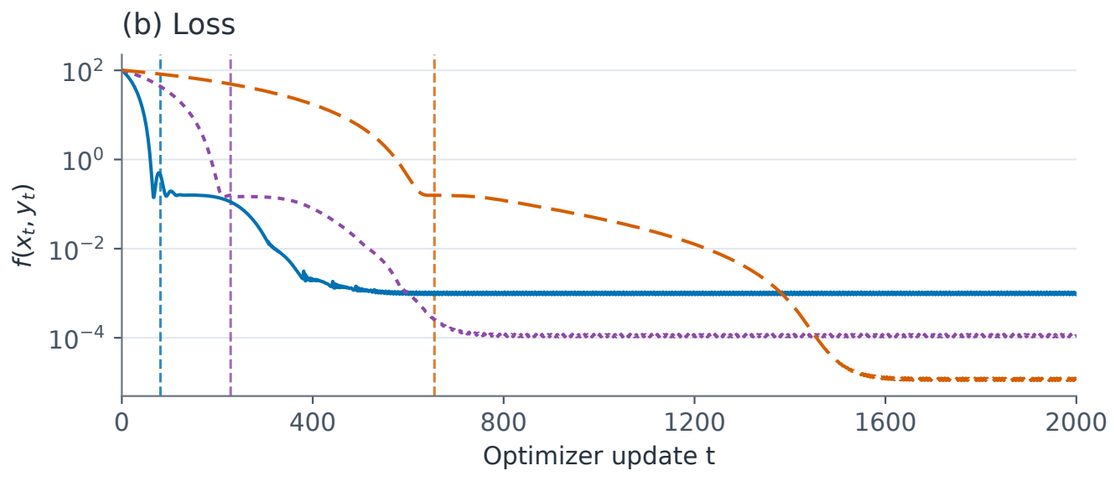
Learning rate · vertical dashed line at first y-turn

$\eta = 10^{-2}$ · turn t = 81 | $\eta = 3.1623 \times 10^{-3}$ · turn t = 228 | $\eta = 10^{-3}$ · turn t = 655

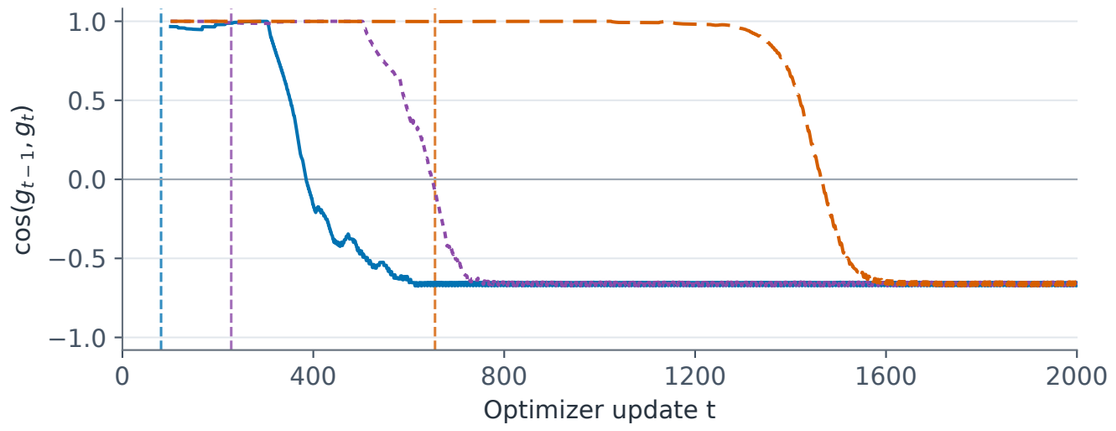
(a) Full optimization trajectories and objective level curves



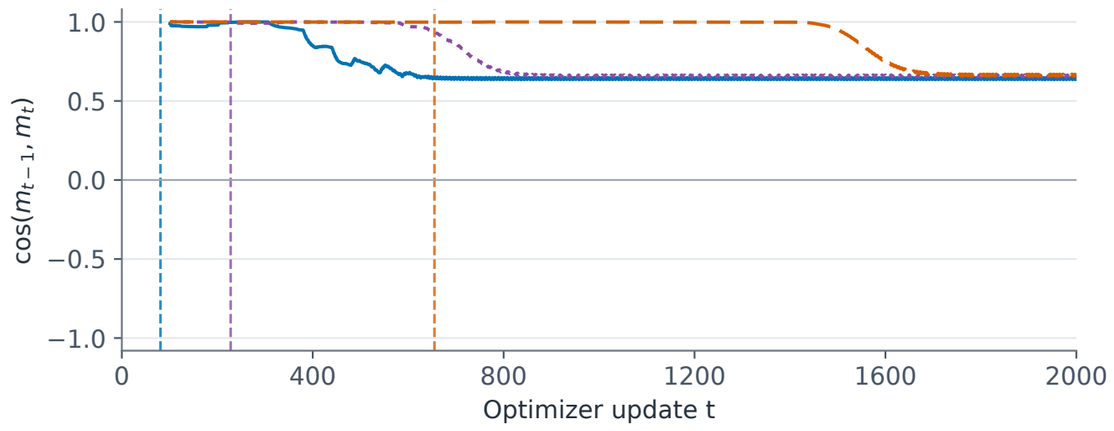
(b) Loss



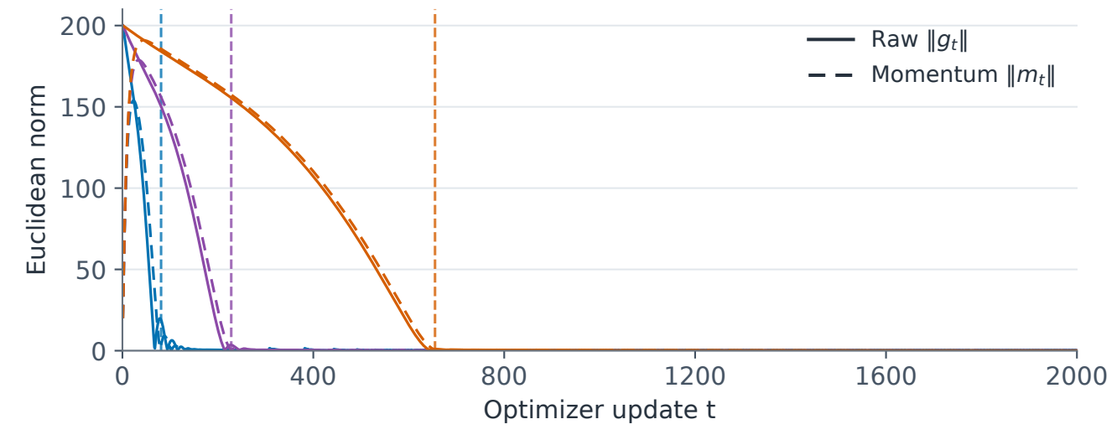
(c) Raw-gradient cosine: 100-step trailing mean



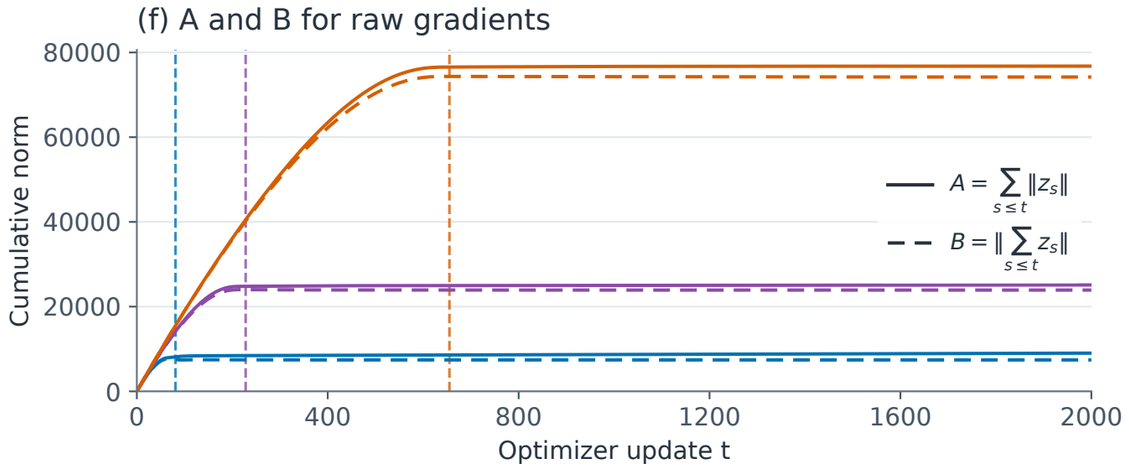
(d) Momentum cosine: 100-step trailing mean



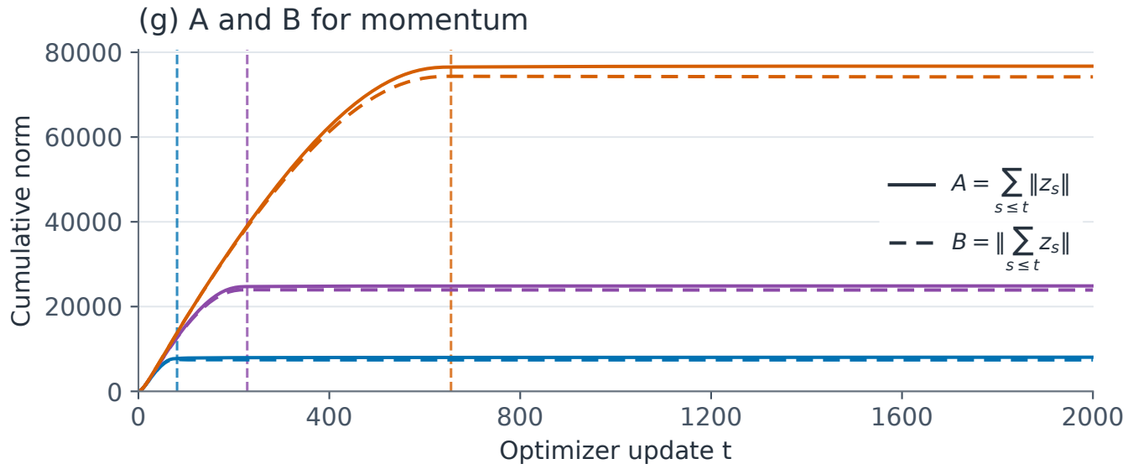
(e) Raw-gradient and momentum vector norms



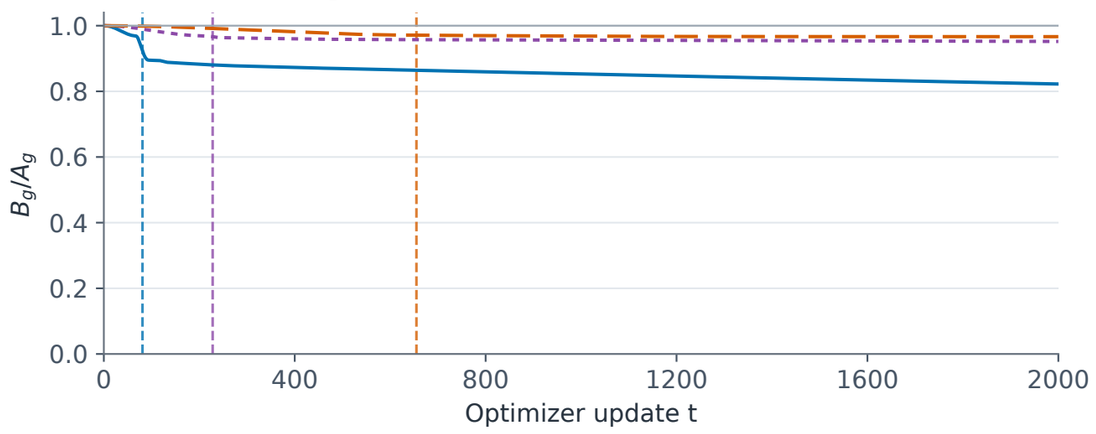
(f) A and B for raw gradients



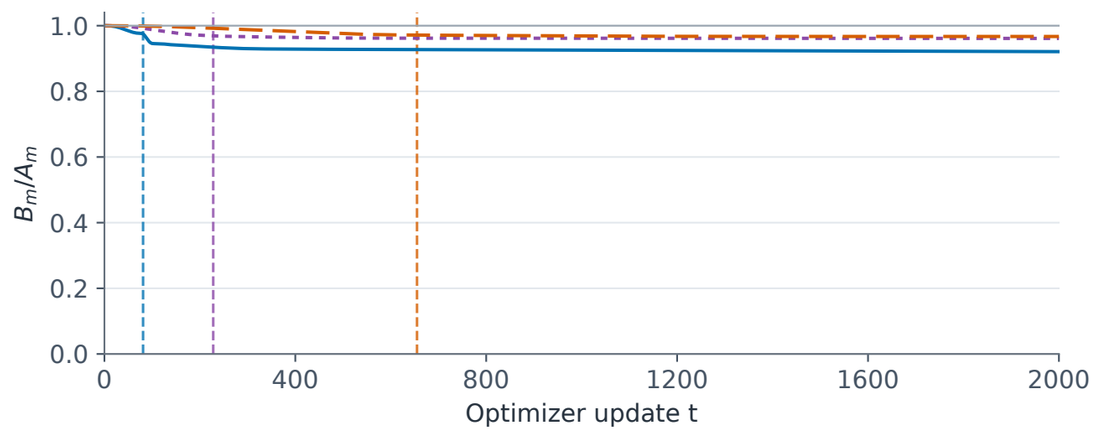
(g) A and B for momentum



(h) B/A for raw gradients



(i) B/A for momentum



$f(x, y) = (1 - x)^2 + 100(y - x^2)^2$. Panels (c,d): 100-step trailing means; complete finite windows only. Other series unsmoothed.

m_t is the stored, uncorrected first moment. Diamonds and LR-colored dashed vertical lines mark the first y-turn.

Vectors at update t use $g_t = \nabla f(x_{t-1}, y_{t-1})$; loss uses (x_t, y_t). First adjacent cosines are undefined. B/A is undefined if A = 0.